Soul-Sync

Project Team

Abdullah Bin Zahid	21I-1388
Muhammad Usman Ghani	21I-2971
Hamza Amin Khokhar	21I-2979

Session 2021-2025

Supervised by

Dr. Mahreen Alam

Department of Computer Science

National University of Computer and Emerging Sciences
Islamabad, Pakistan

June, 2025

Contents

List of Figures

List of Tables

Chapter 1

Introduction

In todays modern era major amount of personal data like thoughts, emotions and expertise is stored in various digital forms, but this data often vanishes into non existence after person's passing. Loved ones and society not only lose emotional attachment but also the knowledge of deceased person. Odem et al. (2010) studies also represent the significance of preservation of digital legacy of a deceased person. Weare currently living in one of the most innovative era for technology in this era almost everyone has a digital footprint including their pictures, videos, textual posts etc. among social media sites like Facebook, Instagram, Twitter (now known as X), Watsapp etc. But despite the treasures of data available there is no concrete way to preserves one's knowledge in a way that we are able to synthetically save one's persona after his death. In other words we lack a tool that has the ability to maintain a digital legacy of late ones. After the death as the time progress that special connection and emotional touch that is unique for everyone fades away if not properly maintained or restored . [1].

1.1 Problem Statement

The purpose of SoulSync is to solve the growing problem of saving and interacting with a person's digital legacy after they die. In today's modern era, most of a person's experi ences, emotions and expertise are shared across various social media platforms and other mediums making it hard for loved ones to access his/her information meaningfully. The connection of the deceased one that could influence future actions and provide comfort will vanish if there is no proper tool for interaction with this knowledge. SoulSync solves this problem by utilizing Artificial Intelligence technologies like Natural Language Pro cessing, Computer Vision, and deep learning models to create an artificial persona that replicates an individual's communication style, wisdom, and emotional resonance allow ing the preservation of their legacy.

1.2 Scope

The SoulSync project is about creating a state-of-the-art AI platform that will produce Artificial Personas based on the analysis of all of a person's digital footprints– his posts on social media, emails, photos, and videos. Sophisticated NLP and CV techniques reproduce the way in which the person communicates, depict emotions, and shares knowledge. One prominent feature is a user-friendly mobile app with which users can feed their data and within hours will be able to generate an interactive Persona, which will respond in text, voice, and visuals. To make it even more natural, the platform develops artificial persona. It will focus on keeping things private and data secure.

1.3 Modules

1.3.1 Legacy Data Aggregation

This module will collect data provided by the user to scrape and categorize the data for further processing

1. Data Organization: Uses artificial intelligence to analyze and store data into their relevant categories automatically.

1.3.2 Data Analysis and Modeling

This module will be used for processing and creation of person's communication style, visual traits and emotional resonance .

1. Textual and Audio Data: It will use textual and audio data for the creation of brain(will have memories in meaningful manner) and recognizing speech style and sentiments.
2. Visual Data: Process Videos and Images for detecting visual context and also for face and speech style creation.

1.3.3 Persona Creation

This module will create dynamic persona.

1. Persona Simulation:Provide user with simulated persona that can interact with user.

2. Communication Interface: Facilitates interaction with digital persona and it will support voice, text and video.

1.3.4 Engagement Module

This module will be responsible for the interaction with digital persona of individual.

1. Interaction: Allows simulated conversation with persona.
2. Emotions and Memory Sharing: Enables user to share personal and emotional memories with persona.

1.4 User Classes and Characteristics

Identify the various user classes that you anticipate will use this product, and describe their pertinent characteristics.

User class	Description
User	The primary users of Soul Sync are the ones that need therapy or knowledge or want to connect with their deceased loved ones.
Admin	The purpose of admin is to control which models can be deployed for public as there can be some controversial models that our system can't detect and also to have a look at Soul Sync users.

Example: User Classes and Characteristics

chapter 1 replace all kardai

Chapter 2

Project Requirements

This chapter describes the functional and non-functional requirements of the project.

2.1 Use-case/Event Response Table/Storyboarding

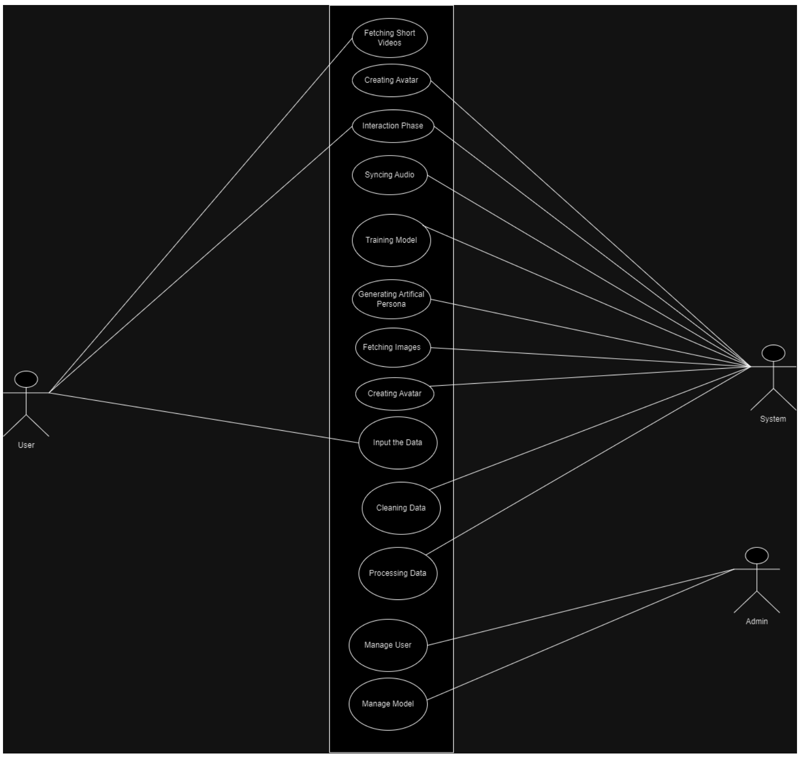

Figure 2.1: Usecase Diagram

UC-01: Input the Data	
Use Case	Input from User
Actors	User
Type	Primary
Description	The user will input the personal data as input for the model. It can be any raw text, image , video and audio.

UC-02: Data Cleaning	
Use Case	Data Cleaning
Actors	System
Type	Primary
Description	The data that is being uploaded to the model will be cleaned out, removing the special character and other non-useful information.

UC-03: Processing Data	
Use Case	Processing Data
Actors	System
Type	Primary
Description	The data that is being cleaned will now be converted into the foam so that it will be given to the model for training,(in technical term embedding,etc.).

UC-04: Fetching Videos	
Use Case	Fetching Videos
Actors	System
Type	Primary
Description	The short video will be given input to the model which is processed frame by frame, image and audio will be extracted.

UC-05: Fetching Images	
Use Case	Fetching Images
Actors	System
Type	Primary
Description	Images that are being feed to model will be used for extracting features like facial expression,etc.

UC-06: Avatar Creation	
Use Case	Avatar Creation
Actors	System
Type	Primary
Description	By using the images data, an avatar will be made that will be an image of user.

UC-07: Training Model	
Use Case	Model Training
Actors	System
Type	Primary
Description	The model will be trained on the input data creating digital persona on the basis of input data.

UC-08: Persona Generation	
Use Case	Persona Generation
Actors	System
Type	Primary
Description	The trained model will now generate the data on its own, based on the pattern it learned in the training.

UC-09: Audio Syncing	
Use Case	Audio Syncing
Actors	System
Type	Primary
Description	By using the audios, the model will replicate the tone and style and will pronounce the new words by same tone.

UC-10: Interaction Phase	
Use Case	Interaction Phase
Actors	System & User
Type	Primary
Description	After the creation and generation of model, now model will be interacting with the user and will answer the questions on the pattern it learned.

UC-11: Managing User	
Use Case	Managing User
Actors	Admin
Type	Primary
Description	Admin will have ability to check on the users, information related to user,data privacy,etc.

UC-12: Managing Model	
Use Case	Managing Model
Actors	Admin
Type	Primary
Description	Admin will have ability to run and execute the model and also restrict the model from generating bad speech and other such things.

2.2 Functional Requirements

2.2.1 Legacy Data Aggregation

1. The system must allow users to add (text,media) type of data
2. The system must categorize data using artificial intelligence

2.2.2 Data Analysis and Modeling

1. The system must process text in a meaningful manner to make the memory bank by storing meaningful data
2. The system must process images and vedios to make persona feel like real by duplicating real person style
3. The system must process audio to store data in memory bank and also to make personas voice.

2.2.3 Persona Creation

1. The system must generate persona based on data processed and analyzed.
2. The system must allow users to interact with artificial persona using userfriendly conversational interface.

2.2.4 Engagement Module

1. The system must allow users to initiate and interact with persona using text and voice
2. The system must enable users to share personal memories for better interaction

2.3 Non-Functional Requirements

2.3.1 Legacy Data Aggregation

1. The aggregation process should be efficient
2. The aggregation process can handle large data within few(1-4) hours

2.3.2 Data Analysis and Modeling

1. The system should support audio and visual media in different formats like MP4 etc
2. The system must ensure privacy for user models

2.3.3 Persona Creation

1. The artificial persona speech should be natural.
2. The artificial persona should provide response in real time with lag time < 1 sec

2.3.4 Engagement Module

1. The artificial personas emotional response should be felt natural to users.
2. The user interface should be user friendly and can be accessible on all mobile device have android version >=8.0

2.4 Domain Model

2.5 Class Diagram

2.6 Sequence Diagrams

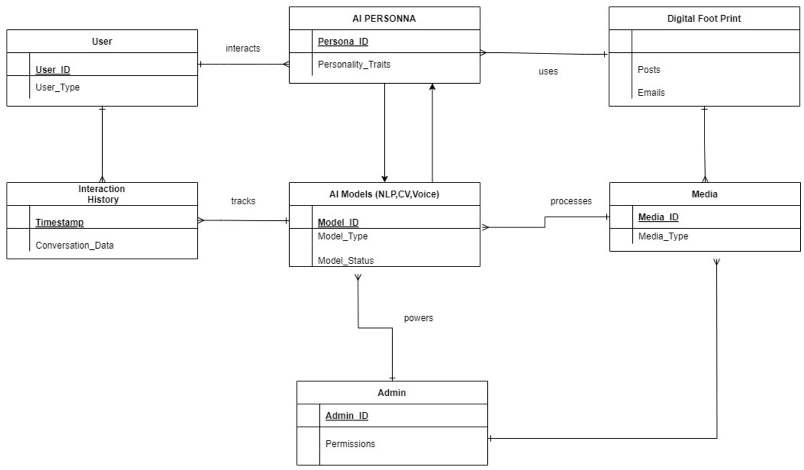

Figure 2.2: Domain Model

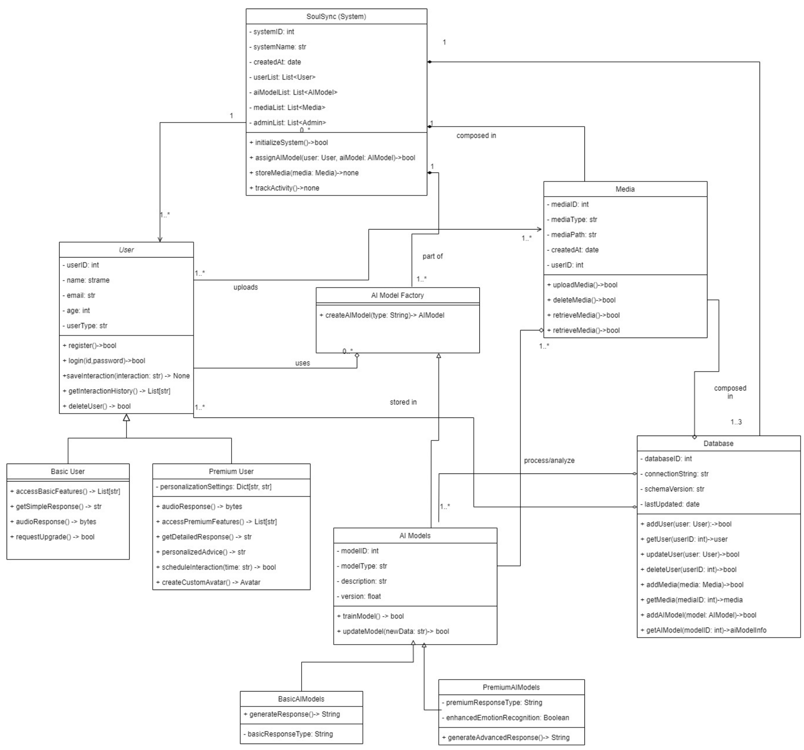

Figure 2.3: Class Diagram

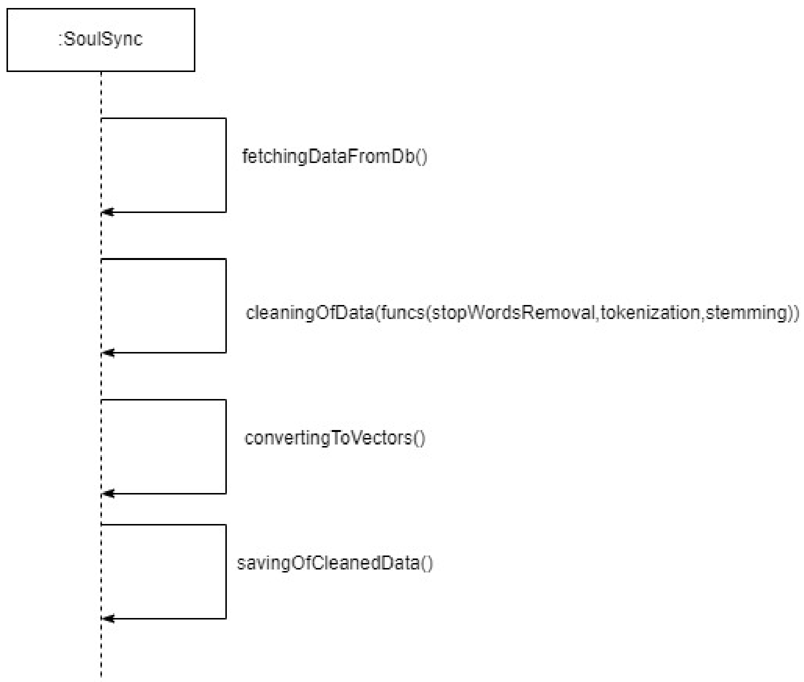

Figure 2.4: Sequence Diagram 1

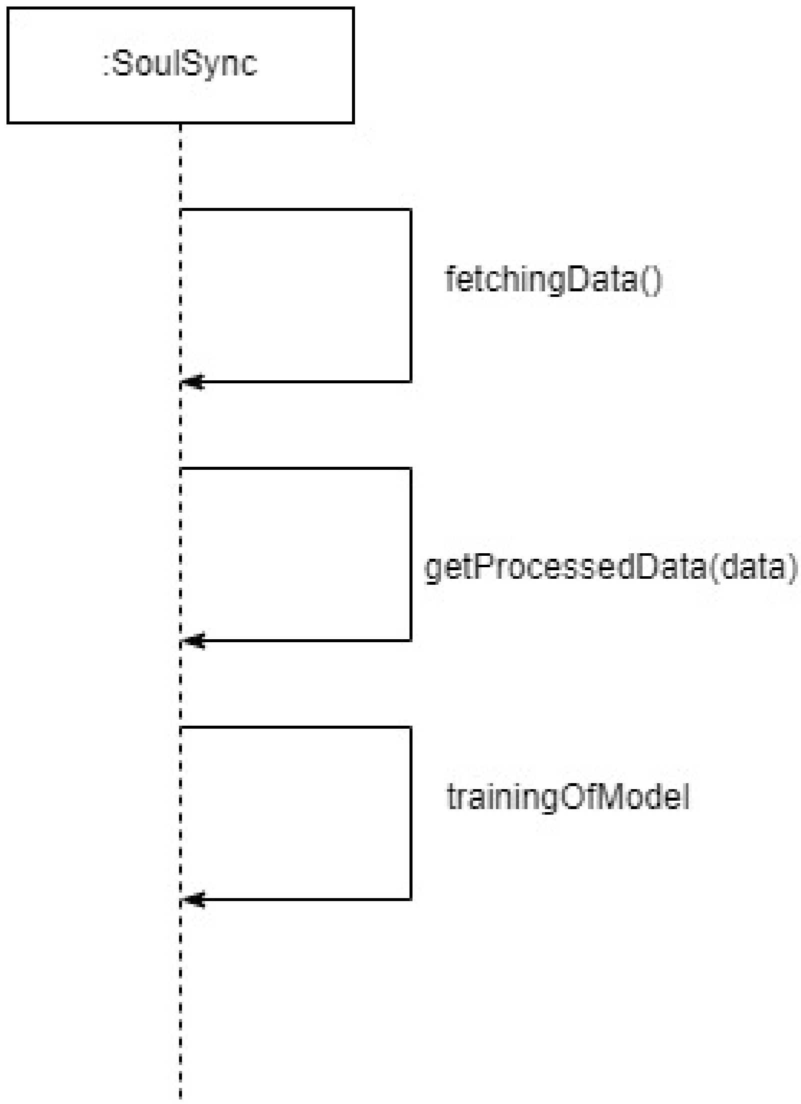

Figure 2.5: Sequence Diagram 2

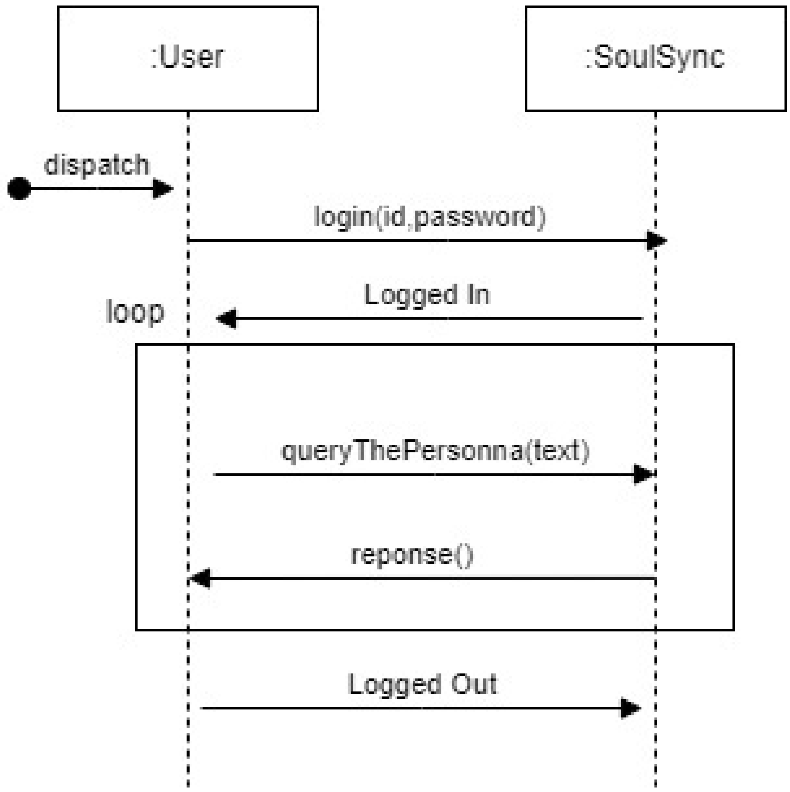

Figure 2.6: Sequence Diagram 3

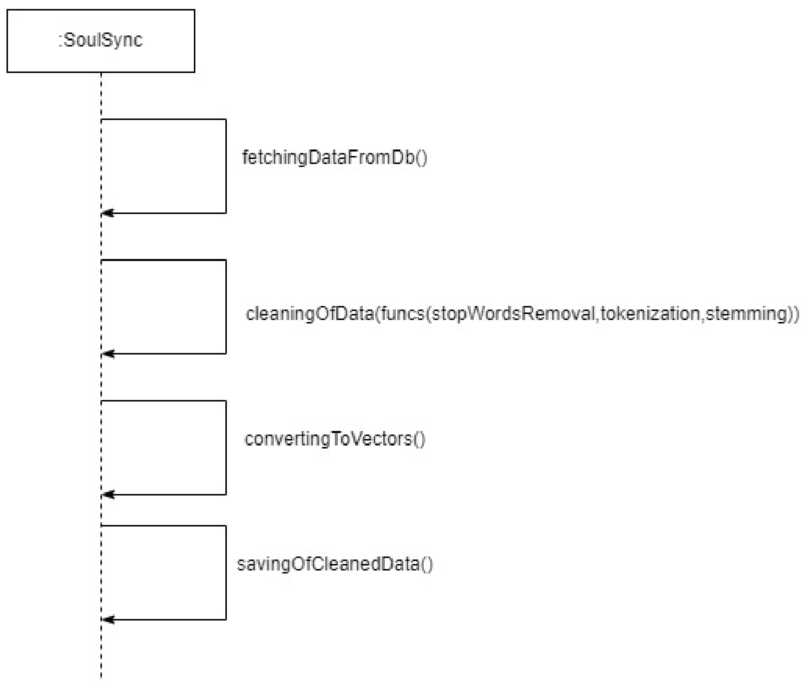

Figure 2.7: Sequence Diagram 4

Chapter 3

System Overview

Give a general description of the functionality, context, and design of your project. Provide any background information if necessary.

3.1 Architectural Design

Develop a modular program structure and explain the relationships between the modules to achieve the complete functionality of the system. This is a high-level overview of how the system's modules collaborate with each other in order to achieve the desired functionality.

Don't go into too much detail about the individual subsystems. The main purpose is to gain a general understanding of how and why the system was decomposed, and how the individual parts work together.

Provide a diagram showing the major subsystems and their connections.

• In initial design stage create Box and Line Diagram for simpler representation of the systems • After finalizing architecture style/pattern diagram (MVC, Client-Server, Layered, Multi-tiered) create a detailed mapping modules/components to each part of the architecture

3.2 Design Models

Create design models as are applicable to your system. Provide detailed descriptions with each of the models that you add. Also ensure visibility of all diagrams.

Design Models for Object Oriented Development Approach

The applicable models for the project using object oriented development approach may include:

• Activity Diagram
• Class Diagram
• Class-level Sequence Diagram
• State Transition Diagram (for the projects which include event handling and backend processes)

Design Models for Procedural Approach

The applicable models for the project using procedural approach may include:

• Activity Diagram
• Data Flow Diagram (data flow diagram should be extended to 2-3 levels. It should clearly list all processes, their sources/sinks and data stores.)
• System-level Sequence Diagram
• State Transition Diagram (for the projects which include event handling and backend processes)

3.3 Data Design

Explain how the information domain of your system is transformed into data structures. Describe how the major data or system entities are stored, processed, and organized.

List any databases or data storage items.

Bibliography

[1] A Kolyshkin and S Nazarovs. Stability of slowly diverging flows in shallow water. *Mathematical Modeling and Analysis*, 2007.